

a Plant context

same accession panel

control



stress 1

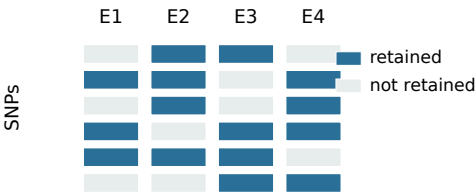


stress 2



accessions measured across environments

b Common SNP × environment grid



complete rows prevent instrument-composition changes between environments

c Covariance-aware estimation

$$r_{jk} = \theta_0 + \theta_1 z_k + \varepsilon_{jk}$$

signed LD



environment correlation



GLS + rank-aware Q

d Plant MR report

